Final Submission Package – Group Software Development Project

Each group must submit the following items as the Final Submission Package. All items should be clearly labeled with the project title and group number.

A. Design & Modelling Artifacts

1. Use-Case Description

- 3–5 main use cases (text descriptions).
- Actors and their responsibilities (e.g. Admin, User, Guest).
- Optional: Simple use-case diagram showing actors and main use cases.

2. UI/UX Design

- Navigation / Screen Flow Diagram showing how windows/pages connect.
- Wireframes or mockups of key screens (login, main dashboard, data entry, search, reports).
- Final screenshots of the implemented UI, with short labels for key components.

3. Database Design

- ER Diagram with entities, attributes, relationships, and cardinalities.
- Relational schema (tables, primary keys, foreign keys).
- Short note on normalization decisions (e.g. designed up to 3NF; removed repeating groups).
- SQL scripts: CREATE TABLE statements and a small sample data INSERT script.

4. OOP / Software Design

- Class diagram showing main classes, attributes, methods, and relationships (inheritance, aggregation, etc.).
- 1–2 sequence or activity diagrams for critical flows (e.g. “Add new item”, “Issue a book”).
- Design note (1–2 pages) explaining how you used:
  - Encapsulation (where and how).
  - Inheritance (which classes, why).
  - Polymorphism (interfaces/virtual methods/overriding, etc.).
- Optional: mention and briefly explain any design patterns used.

B. Implementation Artifacts

5. Source Code

- Full project source code, organised into meaningful folders (e.g. UI, business logic, database, utilities).
- A README file including:
  - Tech stack and version information.
  - Installation and build steps.
  - How to run the application.
  - Default login credentials (if any).

6. Executable / Runnable Build

- Compiled executable or runnable package (or clear instructions to run from source).
- Any configuration files required for database connection and initial setup.

C. Database Functionality Evidence

7. Sample Screens / Reports

- Screenshots that clearly demonstrate:
  - Data entry and validation.
  - Search, filtering, or report generation.
  - Successful creation, update, and deletion of records.

8. Test Queries / Outputs

- 3–5 sample SQL queries that demonstrate meaningful outputs (e.g. top 10 items, overdue items, summary reports).
- Screenshots or exported result tables showing correct query outputs.

D. Testing Artifacts

9. Test Plan & Test Cases

- Short test plan summarizing what was tested (UI, database operations, error handling, boundary cases, etc.).
- A table or list with 5–10 key test cases, each showing:
  - Test case ID and description.
  - Input data.
  - Expected output.
  - Actual result and pass/fail status.
- Screenshots or logs for at least a few passed tests and 1–2 handled error cases.

E. Documentation

10. Final Report

The final report should typically be 10–12 pages and include:

1. Introduction & background.
2. Problem definition & objectives.
3. System requirements (functional and non-functional).
4. System design:
   - Use cases and high-level architecture.
   - UI/UX design.
   - ER diagram and database schema.
   - Class diagram and OOP design explanation.
5. Implementation details and key decisions.

6. Testing strategy and results.

7. Limitations and possible future improvements.

8. Individual contributions of group members.

9. Conclusion.

10. References (if any external resources were used).

11. User Manual

- A short user manual (2–3 pages) that includes:
  - How to install and run the system.
  - Basic workflow: step-by-step instructions for main tasks (with screenshots).
  - How to back up and restore data (if applicable).
  - Known limitations or constraints a user should be aware of.

F. Presentation & Demo

12. Final Presentation Slides

- Slide deck summarizing:
  - Problem and motivation.
  - Overview of features.
  - Key design decisions (UI, database, OOP).
  - Short demo flow (what will be shown in live demo).
  - Team members and their contributions.

13. Live Demonstration

- Live demonstration of the system including:
  - Login or initial access (if applicable).
  - Core operations (adding, updating, searching data).
  - At least one reporting or summary feature.
  - Any additional “nice to have” features implemented.
- All group members should participate in the presentation and/or demo.